Skill Coverage Matrix — AWS AIP-C01

Last updated: **Monday 26 Oct 2026 (after Day 5 + routing mini-lesson)** · Source: official exam guide — 5 domains, 20 tasks, 98 skills

Summary
16 skills covered, 42 partly covered, 40 not yet covered (of 98). Weighted skill coverage (covered = 1, partly = ½, by domain weight): **38%** — a little below the earlier task-level estimate (~42%), because studied tasks still contain uncovered skills. This skill-level figure is the one to trust. Gaps inside already-studied tasks are closed in the gap lesson; the rest are assigned to Days 6–12.

1. Coverage by domain

Domain	Weight	Skills	Covered	Partly	Not yet	Coverage
1. Foundation Model Integration, Data Management, and Compliance	31%	28	7	11	10	45%
2. Implementation and Integration	26%	25	5	11	9	42%
3. AI Safety, Security, and Governance	20%	15	3	4	8	33%
4. Operational Efficiency and Optimization	12%	16	1	9	6	34%
5. Testing, Validation, and Troubleshooting	11%	14	0	7	7	25%

2. Every skill

Skill	What it asks for	Status	Where taught	Gap / plan
1.1.1	Architectures aligned to business needs (FMs, integration patterns, deployment)	½ Partly	D1 RAG vs fine-tuning; D2 agent vs workflow; D5 deployment options	End-to-end design method → Day 7
1.1.2	Proofs of concept on Bedrock to validate feasibility, performance, value	✗ Not yet	—	Day 7
1.1.3	Standard components; Well-Architected Framework + Generative AI Lens	✗ Not yet	—	Day 7
1.2.1	Assess and choose FMs (benchmarks, capabilities, limitations)	✓ Covered	D5 Part 1 (criteria, model evaluation)	—
1.2.2	Switch models/providers without code changes (Lambda, API Gateway, AppConfig)	½ Partly	D5 Converse API; routing lesson (AppConfig map)	Gap lesson: config-driven model switching

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1.2.3	Resilience: Step Functions circuit breaker, CRIS, cross-Region, graceful degradation	½ Partly	D5 cross-region inference, fallbacks	Gap lesson: circuit breaker, graceful degradation
1.2.4	Customisation lifecycle: SageMaker deploy, LoRA/adapters, Model Registry, pipelines, rollback, retire	½ Partly	D5 Part 3 (fine-tuning, CPT, distillation, Custom Model Import)	Gap lesson: LoRA, Model Registry, rollback, retirement
1.3.1	Data validation (Glue Data Quality, Data Wrangler, Lambda, CloudWatch metrics)	✗ Not yet	—	Day 8
1.3.2	Multimodal processing (Bedrock multimodal, SageMaker Processing, Transcribe)	½ Partly	D1 parsing (BDA, Textract); D5 multimodal models	Day 8
1.3.3	Format inputs per model (Bedrock JSON, SageMaker payloads, conversation format)	½ Partly	D2 Converse messages; D5 InvokeModel vs Converse, batch JSONL	Day 8
1.3.4	Improve input quality (Bedrock reformatting, Comprehend entities, Lambda normalisation)	✗ Not yet	—	Day 8
1.4.1	Vector architectures (KB hierarchical, OpenSearch Neural plugin, RDS+S3, DynamoDB for metadata)	½ Partly	D1 KB + vector stores	Gap lesson: OpenSearch Neural plugin, DynamoDB metadata pattern
1.4.2	Metadata frameworks (S3 metadata timestamps, authorship attributes, domain tags)	½ Partly	D1 metadata filtering; PS1 Q3	Gap lesson: .metadata.json attributes
1.4.3	Performance at scale (OpenSearch sharding, multi-index, hierarchical indexing)	✗ Not yet	—	Gap lesson
1.4.4	Integration with document systems, wikis (connectors)	✓ Covered	D1 KB data source connectors	—
1.4.5	Keep vector stores current (incremental, change detection, sync, scheduled refresh)	✓ Covered	D1 sync jobs; PS1 Q5	—
1.5.1	Document segmentation (Bedrock chunking, Lambda fixed-size, hierarchical)	✓ Covered	D1; PS1 Q16	—
1.5.2	Embedding choice (Titan dimensions, domain fit, Lambda batch embeddings)	½ Partly	D1 embeddings; PS1 Q20	Gap lesson: dimension trade-offs, batch embedding
1.5.3	Vector search (OpenSearch, Aurora pgvector, KB managed store)	✓ Covered	D1; PS1 Q2	—
1.5.4	Advanced search (hybrid, rerankers)	✓ Covered	D1; PS1 Q1	—
1.5.5	Query handling (expansion, decomposition, Step Functions transformation)	½ Partly	D1 query decomposition	Gap lesson: query expansion/transformation

Skill	What it asks for	Status	Where taught	Gap / plan
1.5.6	Consistent access (function calling for vector search, MCP clients, standard APIs)	✓ Covered	D2 tools; D3 MCP; PS1 Q17	—
1.6.1	Instruction frameworks (Prompt Management roles, Guardrails, templates)	½ Partly	D5 system prompts; D6 Part 1 guardrails	Day 7
1.6.2	Interactive systems (Step Functions clarification, Comprehend intent, DynamoDB history)	✗ Not yet	—	Day 7
1.6.3	Prompt governance (parameterised templates, approvals, S3 repos, CloudTrail, Logs)	✗ Not yet	—	Day 7
1.6.4	Prompt QA (Lambda output checks, Step Functions edge cases, CloudWatch regression)	✗ Not yet	—	Day 7
1.6.5	Iterative refinement (structured inputs, output specs, chain-of-thought, feedback loops)	✗ Not yet	—	Day 7
1.6.6	Complex prompt systems (Prompt Flows chains, branching, reusable components)	✗ Not yet	—	Day 7
2.1.1	Autonomous systems with memory/state (Strands, Agent Squad, MCP)	✓ Covered	D2, D3; PS1	—
2.1.2	Structured reasoning (Step Functions ReAct, chain-of-thought)	½ Partly	D2 ReAct; D3 Step Functions	Day 7 (chain-of-thought)
2.1.3	Safeguarded workflows (stopping conditions, Lambda timeouts, IAM boundaries, circuit breakers)	½ Partly	D3 Step Functions; PS1 Q29	Gap lesson
2.1.4	Model coordination (specialised FMs, ensembles, selection frameworks)	½ Partly	D3 supervisor/router; routing lesson	Gap lesson: ensembles + aggregation
2.1.5	Human collaboration (Step Functions approvals, API Gateway feedback)	½ Partly	D3 waitForTaskToken; PS1 Q10, Q29	Gap lesson: feedback collection
2.1.6	Reliable tools (Strands custom behaviours, function definitions, Lambda validation)	½ Partly	D2 tool schemas, toolResult errors; PS1 Q21	Gap lesson: parameter validation
2.1.7	MCP servers: Lambda (stateless, light) vs ECS (complex); client libraries	½ Partly	D3 MCP	Gap lesson
2.2.1	Deploy per need (Lambda on-demand, Provisioned Throughput, SageMaker hybrid)	✓ Covered	D5 Part 3	—
2.2.2	LLM-specific deployment (containers, GPU memory, model loading)	✗ Not yet	—	Gap lesson

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2.2.3	Optimised deployment (smaller models, cascading)	✓ Covered	D5 distillation; routing lesson	—
2.3.1	Enterprise connectivity (legacy APIs, event-driven, data sync)	✗ Not yet	—	Day 8
2.3.2	Add GenAI to apps (API Gateway microservices, Lambda webhooks, EventBridge)	✗ Not yet	—	Day 8
2.3.3	Secure access (identity federation, RBAC, least privilege)	½ Partly	D4 IAM/SCP; D3 Identity	Day 8
2.3.4	Cross-environment compliance (Outposts, Wavelength, secure routing)	✗ Not yet	—	Day 8
2.3.5	CI/CD + GenAI gateway (CodePipeline, CodeBuild, tests, rollback, central layer)	✗ Not yet	—	Day 8
2.4.1	Sync/async interaction (Bedrock APIs, SDKs + SQS, API Gateway validation)	✓ Covered	D5 Part 2	—
2.4.2	Real-time (streaming APIs, WebSockets / server-sent events, chunked transfer)	½ Partly	D5 streaming	Gap lesson: WebSockets, SSE
2.4.3	Resilience (SDK backoff, API Gateway rate limiting, fallback, X-Ray)	½ Partly	D5 backoff, fallback	Gap lesson: API Gateway throttling, X-Ray
2.4.4	Model routing (static, Step Functions content-based, metrics, API Gateway)	✓ Covered	Routing mini-lesson (26 Oct)	—
2.5.1	FM API interfaces (streaming via API Gateway, token limits, retries)	½ Partly	D5 Part 2	Day 9
2.5.2	Accessible interfaces (Amplify UI, OpenAPI-first, Prompt Flows no-code)	✗ Not yet	—	Day 9
2.5.3	Business enhancements (Lambda CRM, Step Functions doc processing, BDA)	✗ Not yet	—	Day 9
2.5.4	Developer productivity (Amazon Q Developer)	✗ Not yet	—	Day 9
2.5.5	Advanced apps (Strands, Agent Squad, Step Functions patterns, prompt chaining)	½ Partly	D3	Day 9
2.5.6	Troubleshooting (Logs Insights, X-Ray, Q Developer)	✗ Not yet	—	Day 9 / Day 11
3.1.1	Input safety (Guardrails, custom moderation with Step Functions/Lambda)	½ Partly	D6 Part 1	Day 6

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3.1.2	Output safety (Guardrails, toxicity evaluations, text-to-SQL determinism)	½ Partly	D6 Part 1	Day 6
3.1.3	Reduce hallucinations (KB grounding, confidence scoring, JSON Schema outputs)	½ Partly	D1 grounding; D5 Part 1 Q&A;	Day 6
3.1.4	Defence in depth (Comprehend pre-filter, Guardrails, Lambda post-check, API filtering)	✗ Not yet	—	Day 6
3.1.5	Threat detection (prompt injection, jailbreaks, sanitisation, adversarial testing)	✗ Not yet	—	Day 6
3.2.1	Protected environments (VPC endpoints, IAM, Lake Formation, CloudWatch)	✓ Covered	D4	—
3.2.2	Privacy (Comprehend, Macie, Bedrock privacy, Guardrails, S3 Lifecycle)	✓ Covered	D4	—
3.2.3	Privacy with utility (masking, anonymisation, PII detection)	✓ Covered	D4	—
3.3.1	Compliance (model cards, Glue lineage, source tagging, decision logs)	✗ Not yet	—	Day 9
3.3.2	Data source tracking (Glue Data Catalog, attribution tags, CloudTrail)	✗ Not yet	D4 CloudTrail preview	Day 9
3.3.3	Organisational governance frameworks	✗ Not yet	—	Day 9
3.3.4	Continuous governance (misuse/drift/bias detection, alerting, redaction, logging)	✗ Not yet	—	Day 9
3.4.1	Transparency (reasoning displays, confidence metrics, source attribution, agent tracing)	½ Partly	D1 citations; D2 trace	Day 6
3.4.2	Fairness (fairness metrics, Prompt Management A/B, LLM-as-a-judge)	✗ Not yet	—	Day 6
3.4.3	Policy compliance (Guardrails, model cards, Lambda compliance checks)	✗ Not yet	—	Day 6
4.1.1	Token efficiency (estimation, context pruning, response limits, compression)	½ Partly	D5 max tokens, RAG to shrink input	Day 10
4.1.2	Cost-effective selection (tiered usage, price-performance)	½ Partly	D5; routing lesson	Day 10
4.1.3	Utilisation and throughput (batching, capacity planning, PT optimisation)	½ Partly	D5 Part 3	Day 10

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4.1.4	Caching (semantic, fingerprinting, edge, request hashing, prompt caching)	✗ Not yet	—	Day 10
4.2.1	Responsiveness (pre-computation, latency-optimised models, parallel, streaming)	½ Partly	D5 streaming; PS1 Q30	Day 10
4.2.2	Retrieval performance (index optimisation, query preprocessing, hybrid scoring)	½ Partly	D1 hybrid	Day 10
4.2.3	Throughput (token processing, batch inference, concurrency)	½ Partly	D5 batch	Day 10
4.2.4	Model-specific tuning (parameters, A/B tests, temperature/top-k/top-p)	✓ Covered	D5 Part 1	—
4.2.5	Resource allocation (capacity planning, utilisation monitoring, auto-scaling)	✗ Not yet	—	Day 10
4.2.6	System performance (API profiling, vector query optimisation, latency reduction)	✗ Not yet	—	Day 10
4.3.1	Holistic observability (operational, tracing, business dashboards)	½ Partly	D3 Observability	Day 11
4.3.2	GenAI KPIs (token usage, hallucination rate, drift, invocation logs, cost anomalies)	½ Partly	D4 invocation logging	Day 11
4.3.3	Actionable insights (dashboards, compliance monitoring, forensic audit)	✗ Not yet	—	Day 11
4.3.4	Tool performance (call patterns, tool-calling observability, baselines)	½ Partly	D3 Observability	Day 11
4.3.5	Vector store operations (monitoring, index optimisation, data quality)	✗ Not yet	—	Day 11
4.3.6	GenAI failure modes (golden datasets, output diffing, reasoning path tracing)	✗ Not yet	—	Day 11
5.1.1	Quality metrics (relevance, factual accuracy, consistency, fluency)	½ Partly	D1 RAG metrics	Day 12
5.1.2	Systematic evaluation (Bedrock evaluations, A/B, canary, cost-performance)	½ Partly	D5 model evaluation	Day 12
5.1.3	User-centred evaluation (feedback, ratings, annotation)	✗ Not yet	—	Day 12
5.1.4	QA processes (continuous evaluation, regression tests, quality gates)	✗ Not yet	—	Day 12
5.1.5	Multi-perspective assessment (RAG eval, LLM-as-a-judge, human feedback)	½ Partly	D1; D5; PS1 Q19	Day 12

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5.1.6	Retrieval quality testing (relevance scoring, context matching, latency)	½ Partly	D1 metrics	Day 12
5.1.7	Agent performance (task completion, tool effectiveness, agent evaluations)	✗ Not yet	—	Day 12
5.1.8	Reporting to stakeholders (visualisations, automated reports)	✗ Not yet	—	Day 12
5.1.9	Deployment validation (synthetic workflows, hallucination/semantic drift checks)	✗ Not yet	—	Day 12
5.2.1	Content handling (context overflow, dynamic chunking, truncation analysis)	½ Partly	D5 ValidationException, max_tokens	Day 11
5.2.2	Integration issues (error logging, request validation, response analysis)	½ Partly	D5 error table; D4 KMS/IAM errors	Day 11
5.2.3	Prompt problems (testing frameworks, version comparison, refinement)	✗ Not yet	—	Day 7 / Day 11
5.2.4	Retrieval issues (relevance, embedding diagnostics, drift, chunking fixes)	½ Partly	D1; PS1 Q20 embedding mismatch	Day 11
5.2.5	Prompt maintenance (template testing, Logs, X-Ray, schema validation)	✗ Not yet	—	Day 7 / Day 11